

Calcium

Fact Sheet for Health Professionals

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This is a fact sheet intended for health professionals. For a general overview, see our [consumer fact sheet](#).

Introduction

Calcium, the most abundant mineral in the body, is found in some foods, added to others, present in some medicines (such as antacids), and available as a dietary supplement.

Calcium makes up much of the structure of bones and teeth and allows normal bodily movement by keeping tissue rigid, strong, and flexible [1]. The small ionized pool of calcium in the circulatory system, extracellular fluid, and various tissues mediates blood vessel contraction and dilation, muscle function, blood clotting, nerve transmission, and hormonal secretion [1,2].

Calcium from foods and dietary supplements is absorbed by both active transport and by passive diffusion across the intestinal mucosa [1,3]. Active transport is responsible for most absorption when calcium intakes are lower, and passive diffusion accounts for an increasing proportion of calcium absorption as intakes rise. Vitamin D is required for calcium to be absorbed in the gut by active transport and to maintain adequate calcium levels in the blood [1].

Assessing calcium status

Because almost all calcium in the body is stored in the skeleton, a dual x-ray absorptiometry scan of bone mineral density can assess a person's cumulative calcium status over their lifetime. Total calcium levels can be measured in serum or plasma, but these levels are not a good reflection of an individual's calcium status.

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Recommended Intakes

The Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine has established Recommended Dietary Allowances and Adequate Intakes for calcium. These values range from 1,000 to 1,200 mg for adults and from 200 to 1,300 mg for infants, children, and adolescents, depending on age.

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Sources of Calcium

Food

Milk, yogurt, and cheese contain high amounts of calcium. Some nondairy foods also contain calcium, including certain vegetables (e.g., kale, broccoli, Chinese cabbage) and canned sardines and salmon with bones. In addition, some nondairy foods, such as many fruit juices, milk substitutes, tofu, and cereals, are fortified with calcium. The absorption of calcium varies depending on the type of food; absorption from dairy products and fortified foods is about 30%, while absorption from certain plants may be much lower due to the presence of compounds that form indigestible salts with calcium.

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Dietary supplements

Many dietary supplements contain calcium, usually in the form of calcium carbonate or calcium citrate. The percentage of calcium that is absorbed from

supplements depends on a number of factors, including the form of the calcium and the total amount of elemental calcium consumed at one time.

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Medicines

Certain over-the-counter antacid products contain calcium.

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Calcium Intakes and Status

According to data from the National Health and Nutrition Examination Survey, many people in the United States consume less than the recommended amounts of calcium. Non-Hispanic Blacks and non-Hispanic Asians, as well as people living in poverty, are more likely than other populations to have inadequate calcium intakes.

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Calcium Deficiency

Calcium deficiency can lead to osteoporosis, which causes weak, fragile bones and increases the risk of falls and fractures. It can also cause rickets in children and osteomalacia in children and adults, although these conditions are more commonly caused by vitamin D deficiency.

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Groups at Risk of Calcium Inadequacy

Certain groups of people are more likely than others to have calcium inadequacy. These include postmenopausal women and individuals who avoid dairy products.

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Calcium and Health

This section focuses on six health conditions and diseases in which calcium might play a role: bone health in older adults, cancer, cardiovascular disease (CVD),

preeclampsia, weight management, and metabolic syndrome.

Bone health in older adults

Age-related bone loss can lead to osteoporosis and an increased risk of bone fractures. The U.S. Food and Drug Administration has approved a health claim for the use of supplements that contain calcium and vitamin D to reduce the risk of osteoporosis in older adults. However, not all clinical trials have found that these supplements improve bone health or reduce the risk of fractures in this population.

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Cancer

Some evidence suggests that calcium may reduce the risk of certain types of cancer, although the relationship between calcium intake and cancer risk remains unclear. Specifically, some observational studies have reported an association between higher calcium intakes and a lower risk of colorectal cancer, but these findings are not always corroborated by the results of calcium supplementation trials. Consuming high levels of calcium from dairy may be associated with a higher risk of prostate cancer; however, the studies that have evaluated the effects of calcium on the risks for other cancers, such as ovarian or breast cancer, have had mixed results.

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Cardiovascular disease

According to several large observational studies, people with lower calcium intakes may have an increased risk of hypertension, stroke, and atherosclerosis. However, some prospective cohort studies and randomized controlled trials have found that the use of calcium supplements might actually increase the risk of cardiovascular disease, while other studies have found no link between calcium supplements and cardiovascular disease risk or outcomes. More research is needed to clearly define the associations between dietary calcium and calcium supplements and cardiovascular disease.

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Preeclampsia

Preeclampsia is a leading cause of maternal and neonatal morbidity and mortality in the United States. Calcium supplements might reduce the risk of preeclampsia in pregnant women who have inadequate calcium intakes.

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Weight management

Whether dietary or supplemental calcium intakes affect weight management is unknown. While some studies have reported a lower prevalence of overweight or obesity or a lower risk of gaining weight among people with higher calcium intakes, others have found no link between calcium intakes and factors such as body mass index, body weight, or body fat.

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Metabolic syndrome

A few analyses and one clinical trial have indicated that higher calcium intakes may reduce a person's risk of developing metabolic syndrome, especially in women, but more evidence is needed to clarify this potential link.

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Health Risks from Excessive Calcium

Higher intakes of supplemental calcium might increase the risk of kidney stones. According to some research, calcium supplements have the potential to increase the risk of cardiovascular disease. Higher calcium intakes might also increase the risk of prostate cancer. The tolerable upper intake level for calcium ranges from 2,000 mg to 2,500 mg for adults and from 1,000 mg to 3,000 mg for infants, children, and adolescents, depending on age.

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Interactions with Medications

Calcium supplements may interact with medications, and some medications may affect calcium levels. These medications include dolutegravir, levothyroxine,

lithium, and quinolone antibiotics.

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Calcium and Healthful Diets

In general, a person's nutritional needs should be met primarily through the diet, including fortified foods. Dietary supplements may be useful in cases where it is not possible to meet the needs for specific nutrients through food alone, especially during certain life stages. The *Dietary Guidelines for Americans* offers a general description of healthy dietary patterns.

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